AP Precalculus

Unit 2A Day 7 Practice

Graphing Sine Functions

1. Graph: $f(x) = 4\sin\left(\frac{2}{3}x - \frac{\pi}{9}\right) + 2$
   - Amplitude
   - Period
   - Phase shift
   - Vertical shift
   - Domain
   - Range

2. Graph: $f(x) = -2 + \frac{1}{2}\sin\left(\frac{\theta}{3} + \frac{2\pi}{3}\right)$
   - Amplitude
   - Period
   - Phase shift
   - Vertical shift
   - Domain
   - Range

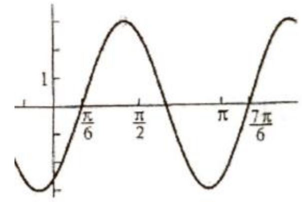
3. Graph: $f(x) = 3\sin(4\theta) - 1$
   - Amplitude
   - Period
   - Phase shift
   - Vertical shift
   - Domain
   - Range

4. Graph $f(x) = 4sin(3x - 6\pi) + 1$
   - Amplitude
   - Period
   - Phase shift
   - Vertical shift
   - Domain
   - Range

5. Fill out the characteristics. Then, write a sine equation to represent the functions graphed.

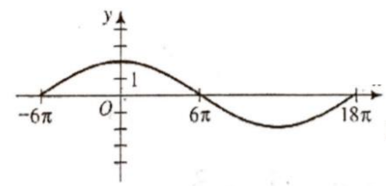
a. Amplitude _________ Period __________

Phase shift $\frac{\pi}{6}$ Vertical shift __________



c. Amplitude __________ Period __________

Phase shift 6π Vertical shift __________



6. Use the given characteristics to write an equation.

a. Sine curve with amplitude $= \frac{1}{2}$, period $= \frac{\pi}{2}$, phase shift = none, and vertical shift = up 7

b. Sine curve with amplitude $= \frac{2}{3}$, period $= \pi$, phase shift $= -\frac{\pi}{4}$, and vertical shift = none.